

Erajaya Value-Added Services

Odoo Hub — Centralized ERP Platform for Erajaya Group

Internal Business Presentation · Product Owner VAS · Mei 2026

Executive Summary

Odoo Hub: platform terpusat untuk percepatan delivery ERP lintas vertikal Erajaya Group.

- Centralized Module Repository — 1 sumber kebenaran modul, reusable lintas vertikal
- Simplified Deployment — provisioning tenant baru otomatis via orchestrator
- Centralized Monitoring — 1 control plane untuk seluruh tenant operasional
- Mandays implementasi turun signifikan karena modul standar siap pakai

82

Modul Siap Pakai

~70%

Modul Reusable

~77

Mandays / Tenant

99.5%

SLA Target

Hub Platform Stack

AI Layer

ai-gateway (Claude / OpenAI / Ollama) · Ask-AI, anomaly inbox, NLQ, OCR receipt, churn prediction

Observability Plane

Prometheus · Grafana · Loki · Alertmanager · custom-predictor (capacity forecast lintas tenant)

Multi-Tenant Runtime

Odoo 19 + 82 custom modules · DB-per-tenant · Caddy/nginx + TLS · Redis · object storage

Tenant Orchestrator

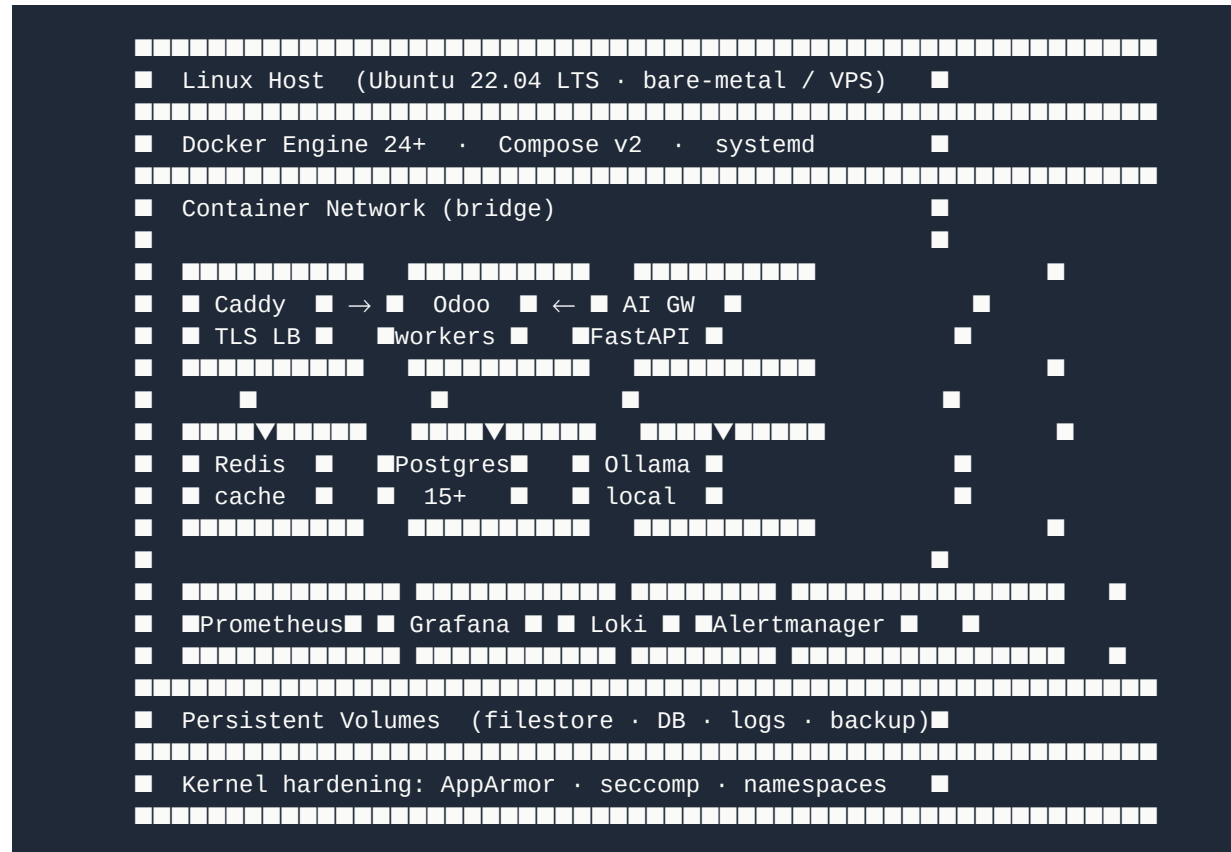
FastAPI service: SSH bootstrap, Docker stack, DB create, module install, mail config — semua otomatis

Centralized Module Repository

Single source of truth 82 modul (core · ee-gap · localized · ops · vertical template) · LGPL-3 · CI-tested

High-Level Architecture (Linux-based)

Stack berjalan di atas Linux host dengan Docker — open, portable, dan tanpa lock-in OS.



Key Decisions

- OS: Ubuntu 22.04 LTS — 5 tahun security update
- Runtime: Docker 24+ + Compose v2, non-root user di tiap container
- Image hardening: distroless / Alpine base, CIS-aligned
- Storage: persistent volumes (filestore, DB, log, backup) bind-mount
- Network: bridge internal; hanya Caddy expose port 80/443
- Firewall: UFW / iptables — port 22 (SSH ops VPN), 80, 443 saja
- TLS: Caddy ACME (Let's Encrypt) auto-renew
- Patch: unattended-upgrades untuk security patches OS
- Backup: pg_dumpall nightly + filestore rsync ke object storage
- Portable: stack sama untuk on-prem, VPS, atau cloud (AWS/GCP/Azure)

Centralized Module Repository — Reuse Matrix

Modul kunci dipakai ulang lintas 6 vertikal Erajaya — bukan develop ulang per tenant.

Module	F&B;	Active	Eraspac	Distrib	Service	Corp
custom_core	✓	✓	✓	✓	✓	✓
custom_accounting_full	✓	✓	✓	✓	✓	✓
custom_hr_payroll_id	✓	✓	✓	✓	✓	✓
custom_attendance	✓	✓	✓	✓	✓	✓
custom_approval_engine	✓	✓	✓	✓	✓	✓
custom_pdp_* (6 modul)	✓	✓	✓	✓	✓	✓
custom_coretax + bupot	✓	✓	✓	✓	✓	✓
custom_helpdesk	✓	✓	✓	✓	✓	✓
custom_whatsapp	✓	✓	✓	✓	✓	✓
custom_pos_id	✓	✓	✓	—	—	—
custom_ecommerce	✓	✓	✓	—	—	—
custom_wms_* (3 modul)	—	✓	✓	✓	✓	—
custom_field_service	—	—	—	—	✓	—
custom_subscription	—	—	—	—	—	✓

Note: ~70% modul shared lintas vertikal · ~30% extension vertical-specific. Modul baru ditambahkan sekali di repo → langsung tersedia untuk semua tenant.

Module Library — Capability Highlights

Finance & Tax (Indonesian-localized, ready)

PSAK 5-digit CoA · Intercompany & consolidation · Fixed asset depreciation · PPh 21 TER · BPJS Kes/TK · SPT 1721 A1 · PPN DPP Nilai Lain · e-Faktur Coretax · Bupot Unifikasi — semua built-in, tidak perlu develop per vertikal

Sales · CRM · Commerce

Predictive lead scoring · Drip campaigns · Mass email · Midtrans/Xendit/DOKU · JNE/JNT/SiCepat/AnterAja · Subscription MRR · Asset rental w/ BAST · WhatsApp QR event ticket

Manufacturing & WMS

MRP PLM (ECO + BoM versioning) · Quality + CAPA · Maintenance MTBF/MTTR · WMS putaway/cycle-count/to-engine · Mobile barcode · Zebra HHT PWA · IoT webhook

Human Capital (Indonesian-localized, ready)

Geofence attendance · Cuti UU Cipta Kerja · Performance appraisal 360 · Recruitment + job-board webhook · Expense OCR receipt · Billable timesheet -> payroll

Service Operations

Helpdesk SLA + escalation · Field Service dispatch · Repairs w/ warranty · Appointments · Livechat -> AI suggested reply · Frontdesk visitor

Productivity & Cross-Cutting

Studio-Lite (no-code customization) · Dashboards KPI + AI NLQ · Spreadsheet · Documents · E-signature · Knowledge wiki · Generic approval engine (delegation + OOO + SLA escalation)

Indonesian Localization Ready

Aturan akunting, perpajakan, ketenagakerjaan, dan data protection Indonesia — built-in, package siap pakai per tenant.

Domain	Cakupan Localized
Akunting (PSAK)	Chart of Accounts 5-digit aligned PSAK · Intercompany automation · Consolidation + eliminations · Fixed asset depreciation · Faktur Pengganti workflow
Perpajakan (DJP)	e-Faktur Coretax (NSFP 17 digit PER-11/PJ/2025) · Bupot PPh 21/23/26/Unifikasi · PPh 21 TER (PP 58/2023) · PPN DPP Nilai Lain (PMK 131/2024) · Sertel Fernet-encrypted · Pajakku ASPP H2H adapter
HR & Ketenagakerjaan	BPJS Kesehatan & Ketenagakerjaan (JHT/JKK/JKM/JP) · PTKP & THR · SPT 1721 A1 · Cuti UU Cipta Kerja · Payslip approval flow
Data Protection (UU PDP)	Klasifikasi data field-level · Consent management · DSAR endpoint · Audit log append-only hash-chained · PII masking · Retention auto-purge

Note: Pendekatan adapter pattern: ASP/regulasi berubah → swap implementasi tanpa ubah workflow tenant.

Simplified Deployment — Tenant Orchestrator

Provisioning tenant baru otomatis, estimasi ~30 menit dari permintaan ke siap UAT.

Hub-Portal (Vite + React 18 · port 18000)
Operator klik 'Provision Tenant'

Tenant Orchestrator (FastAPI)

tenant1 DB (isolated) tenant2 DB (isolated) tenantN DB (isolated)

Shared plane:

- ai-gateway • Postgres cluster • Redis
- Object storage (filestore, sertel)
- Observability (Prom/Grafana/Loki)

Key Decisions

- Step 1 — SSH bootstrap target VPS (Docker + Caddy install)
- Step 2 — Pull stack: Odoo + Postgres + Redis + module repo
- Step 3 — Create database tenant + apply addons path
- Step 4 — Install modul standar (sesuai profile vertikal)
- Step 5 — Generate Caddy route + TLS otomatis (ACME)
- Step 6 — Konfigurasi mail (SMTP / IMAP) + integrasi (Pajakku, payment)
- Step 7 — Smoke test + handover ke PO untuk UAT
- Total: ~30 menit / tenant — tidak perlu DevOps mandays manual

Centralized Monitoring

Apa yang Dimonitor	Manfaat Operasional
● Odoo runtime metrics per tenant (requests, latency, error rate) ● Database health (connections, slow query, replication lag) ● AI gateway cost & latency per tenant ● Pajakku ASPP circuit state (open / closed / half-open) ● Audit chain integrity (PDP hash-chain verifier nightly) ● Tenant resource usage (CPU, memory, disk, filestore size) ● Capacity forecast 7-hari via custom-predictor ● TLS expiry, backup status, scheduler health	● 1 dashboard Grafana untuk N tenant — bukan login per server ● Alert centralized via Alertmanager → ops on-call ● Proactive scaling — predictor rekomendasi upgrade hardware sebelum bottleneck ● MTTR turun — runbook + log + metric satu tempat ● Tenant SLA visible — laporan uptime per bulan otomatis ● Cost attribution per tenant (AI, storage, compute) ● Audit-ready: alert log + immutable trail

Stack: Prometheus (scrape 15s) · Grafana · Loki · Alertmanager · custom-predictor sidecar.

Security Posture

Application & Data Isolation

- DB-per-tenant isolation — bukan schema-per-tenant
- RBAC: Odoo groups + record rules per modul
- Multi-tier approval w/ delegation, OOO, SLA escalation
- Append-only audit log + PostgreSQL trigger
- Tenant allow-list per request (HMAC-validated)
- Secrets via SOPS-encrypted di repo + Fernet for sertel

Infrastructure & Pipeline

- CIS-hardened distroless containers, non-root
- AppArmor + seccomp profiles
- TLS termination + HSTS (Caddy / nginx)
- Pre-commit: gitleaks, ruff, bandit, hadolint
- CI: Semgrep (SAST) + pip-audit + Trivy + cosign signing
- Nightly pg_dumpall + DR runbook (drill executed Q2 2026)

AI Layer — Tertanam di Seluruh ERP

Infrastruktur AI

- ai-gateway (FastAPI sidecar) — multi-provider abstraction
- Provider switch via env: Claude / OpenAI / Ollama (local)
- HMAC-validated Odoo -> gateway calls
- Prompt caching untuk efisiensi cost
- Per-tenant rate limit & quota
- custom-predictor — tabular ML, capacity forecasting 7-hari

Fitur AI di Modul Bisnis

- Ask AI server-action di 9 model utama (invoice, payslip, picking)
- Anomaly Inbox — scan harian + severity + suggested action
- NLQ Chat — query natural language dengan PDP masking
- Document Auto-Classify — saran tag + classification
- AI churn prediction (subscription)
- AI suggested reply (livechat), moderation (forum)
- AI OCR receipt (expense), task breakdown (todo)
- Predictive lead scoring (CRM), spreadsheet helpers

Business Value — Implementation Mandays per Tenant

Estimasi standar onboarding 1 vertikal baru di Odoo Hub dengan semua modul standar siap pakai. Angka dalam tanda kurung = mandays opsional (hanya jika ada custom development).

Implementation Phase	ITBP	PO	DevOps	Developer (opt)
Requirement gathering / BRD	5	8	—	—
Tenant provisioning (orchestrator)	—	1	2	—
Module activation (standard)	3	5	2	—
Master data setup (CoA, partner, employee)	5	3	—	—
Configuration (RBAC, approval, workflow)	3	5	1	—
Integration setup (Pajakku, payment, WA)	2	2	3	(2)
Custom module development	—	(3)	—	(15)
UAT + bug-fix	5	5	1	(5)
Training	3	5	—	—
Go-live + hypercare	3	3	2	—
Subtotal — tanpa custom dev	29	37	11	0
Subtotal — dengan custom dev	29	40	11	22

Note: Standar: ~77 mandays/tenant (lintas 3 role). Dengan 1 custom module: ~102 mandays. Baseline implementasi Odoo tradisional tanpa hub: ~200-300 mandays. Saving 60-70% per onboarding setelah library modul ready.

VAS Productization — 5 Product Lines

Platform Hub membuka monetisasi ke ekosistem Erajaya & eksternal.

VAS Product	Modul Penopang	Target Market
ERP-as-a-Service Multi-Tenant	Hub stack lengkap + orchestrator	Internal tenants & afiliasi Erajaya
Localized Compliance Bundle	custom_coretax* + Pajakku + custom_pdp_* + custom_hr_payroll_id	UMKM ekosistem Erajaya (Eraspaces partners, F&B; franchisee)
AI Operations Layer	ai-gateway + custom_ai_features + predictor	Bundled tiap tenant; advanced tier opt-in
HHT / Field Ops Bridge	custom_hht_bridge (Zebra PWA) + field_service	Service Center, distribution, warehouse
Vertical Template Accelerator	Repo modul + onboarding journey	Vertikal baru di dalam Erajaya Group

Roadmap

Q2 2026 — DONE / IN PROGRESS

- 82 modul base shipping
- Multi-tenant orchestrator
- Pajakku ASPP adapter
- Hub-Portal UI (Vite+React)
- Production drill #1 executed

Q3 2026

- Onboard 3 internal vertical (F&B;, Service Center, Corp Svcs)
- Centralized monitoring dashboard rollout
- HHT rollout untuk service vertical
- AI cost optimization (cache hit target 60%)

Q4 2026

- Onboard tenant 4–10 (target Y1)
- Module reuse audit + dedup
- ESG report otomatis (POJK 51/2017)
- Marketplace add-on (3rd-party vertical modules)

2027

- Tenant 11–25 (post-ROI threshold)
- Open partner program (ASPP, integrator)
- Localized Compliance Bundle GTM eksternal

Risks & Mitigations

Risk	Mitigation
Odoo 19 upstream upgrade breaks customs	custom_studio_lite declarative, modules pinned 19.0.x.y, CI per release
Pajakku API / Coretax submission gagal	Circuit breaker + manual fallback Coretax portal + ops alert
Tenant data leakage	DB-per-tenant + HMAC + tenant allow-list di ai-gateway
AI cost explosion	Per-tenant quota + prompt cache + local Ollama fallback
Mandays meleset dari estimasi karena scope creep	BRD freeze sebelum provisioning · change request via approval engine · re-baseline per onboarding

Call to Action

Untuk Decision Maker Erajaya — endorsement & pilot funding.

1

Endorsement

Odoo Hub sebagai backbone delivery ERP Erajaya Group

2

Pilot Funding

3 vertikal pertama (F&B, Service Center, Corporate Services) — Q3 2026

3

Mandays Baseline Calibration

Lock estimasi mandays per vertikal selama 2 onboarding pertama untuk kalibrasi

4

Hiring Runway

2 senior backend, 1 SRE, 1 PO untuk scale Y1 -> Y2

5

GTM Eksternal

Localized Compliance Bundle sebagai produk eksternal Q4 2026

Terima Kasih

~~Terima Kasih~~

Diskusi & Q&A;

Product Owner — Value-Added Services · Erajaya · Mei 2026